SPECIFICATION — NODE 5 AS SYSTEM CONTEXT ANCHOR — PAGE 2

UPPER-BODY CONTEXT DEVICE ARCHITECTURE, DERIVED SYSTEMIC STATE, AND INTIMATE-SIGNAL INTERPRETATION CONTROLS (NON-LIMITING)

A. Overview (Non-Limiting)

In various embodiments, Node 5 is an upper-body wearable configured to capture systemic physiologic context required to interpret intimate-region signals produced by Nodes 1–4. Node 5 is not required to be a novel wearable in isolation; rather, its technical role within the disclosed system is to provide systemic state determination, artifact detection, and trust scoring that intimate-region devices cannot reliably obtain locally.

B. Exemplary Form Factors and Placement (Non-Limiting)

In various embodiments, Node 5 is implemented as one or more of: 1. chest-mounted device (including patches, straps, or chest modules) providing stable ECG and respiration-adjacent motion patterns; 2. ring providing stable optical coupling for PPG and strong wear integrity indicators; 3. wristband providing PPG, EDA, temperature, and motion sensing with continuous wear detection; and/or 4. other upper-body form factors capable of producing substantially similar context outputs. In various embodiments, multiple form factors may be offered as equivalents wherein the system consumes Node 5 outputs as context tags and indices, not raw waveforms, thereby allowing hardware substitution without changing system function.

C. Node 5 Sensor Suite (Non-Limiting)

In various embodiments, Node 5 includes one or more sensors and interfaces including, without limitation: 1. ECG electrodes for cardiac electrical sensing; 2. PPG optical module (emitter/receiver) for pulse acquisition; 3. 6-axis IMU for motion, posture/orientation, and artifact detection; 4. respiratory dynamics sensing (derived from chest motion patterns, impedance-based respiration, or equivalent); 5. temperature sensor(s) for thermal context and coupling plausibility; 6. EDA/skin conductance electrodes for autonomic activation context; and 7. wear/contact integrity sensing, including electrode impedance checks, optical coupling checks, capacitive proximity, and/or contact integrity heuristics.

D. Derived Systemic State Outputs (Non-Limiting)

In various embodiments, Node 5 generates one or more derived outputs used by the system to interpret intimate signals, including: 1. Activity/Rest State: a classification of rest vs. active vs. high-motion periods derived from IMU and physiologic plausibility checks; 2. Autonomic Activation Index: a confidence-scored indicator derived from HRV features and/or EDA features; 3. Thermal Context Index: a temperature-based tag used to interpret vascular and tissue-related signals and to support drift plausibility checks; 4. Respiratory Context Index: respiration rate/variability outputs when available, used to contextualize physiologic state; 5. Wear Integrity Index: a confidence indicator that the device is being worn correctly with adequate coupling; and 6. Composite Context Vector: a combined representation used for gating and normalization. In various embodiments, these outputs are computed per-window and stored as privacy-preserving indices and tags rather than raw waveforms.

E. Why Intimate Nodes Alone Are Insufficient Without Node 5 (Non-Limiting)

In various embodiments, Nodes 1–4 can capture intimate-local signals, but do not reliably capture the whole-body conditions necessary to interpret those signals. Non-limiting reasons include: 1. state dependence of vascular/tissue signals on systemic stress/activation and temperature; 2. motion and posture artifacts that are difficult to infer solely from intimate-local sensors (particularly during sleep or daily life); 3. coupling ambiguity, where local signals may degrade without clear indication of whether changes are physiologic vs. contact-related; and 4. lack of a systemic reference baseline for longitudinal comparability. Accordingly, Node 5 provides a systemic interpretation layer that reduces false inference and improves time-to-time comparability of intimate outputs.

F. Node 5 as an Interpretation Control Layer for Nodes 1–4 (Non-Limiting)

In various embodiments, Node 5 context outputs are consumed as interpretation controls that alter how Node 1–4 data is accepted, weighted, normalized, or displayed. Non-limiting control behaviors include: 1. window suppression (exclude windows where Node 5 indicates high motion or poor wear integrity); 2. confidence weighting (down-weight low-confidence windows instead of excluding); 3. context normalization (compare only windows sharing similar systemic context, e.g., rest-to-rest, sleep-to-sleep); 4. context tagging (label intimate outputs with systemic tags, e.g., elevated activation, thermal drift possible, poor wear integrity); and 5. repeatability scoring (evaluate whether changes persist across similar contexts rather than across mixed contexts).

G. Examples of Systemic Context Conditioning (Non-Limiting)

1. Nocturnal patterns (Node 1): Node 5 classifies sleep/rest windows and suppresses segments with motion artifacts, improving reliability of overnight pattern interpretation. 2. Longitudinal intravaginal physiology (Node 2): Node 5 provides activation and thermal context so impedance/hydration/pH-related trends are not misread during systemic stress or temperature drift. 3. Self-scan geometry (Node 3): Node 5 provides stability and wear integrity context for repeatability scoring and quality labeling of scan sessions. 4. Stimulus-response sessions (Node 4): Node 5 motion gating improves compliance/contact characterization and reduces false responses caused by movement or poor systemic stability.

H. Variations

Any embodiment described herein may omit sensors, add sensors, substitute equivalent modalities, alter form factor, and/or alter processing distribution while maintaining the functional intent of Node 5 providing upper-body context required to interpret intimate-region data as state-conditioned, reliability-scored physiology.